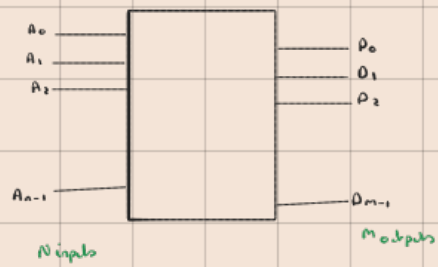


Decoder



$$M = 2^N$$

Applications of Decoder

1) Code Conversion

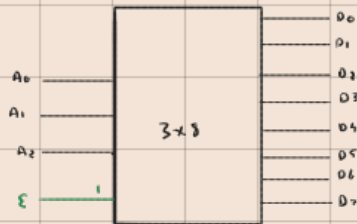
- Using a 3×8 Decoder to convert 3 bit binary into octal from 0 - 7
- Using a 4×16 Decoder to convert BCD into Decimal.
 - Note that only 1 - 9 output pins are used.

2) Address Decoding

3) Generation of Control Signals

4) Logic Circuit Implementation

3x8 Decoder



∴ May also have an Enable Input.

Can be Active High or Active Low

- For Active High

- Signal high means enabled

- Signal low means no output

- For Active Low

- Signal low, Enable

- Signal High means no output

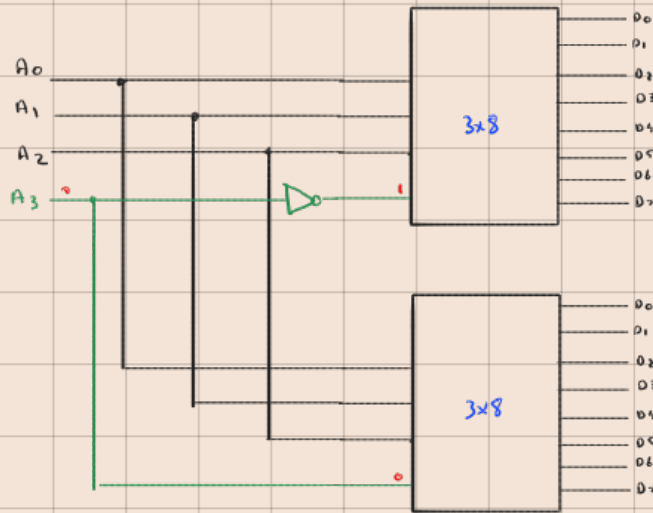
A_0	A_1	A_2	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7
0	0	0	1							
0	0	1		1						
0	1	0			1					
0	1	1				1				
1	0	0					1			
1	0	1						1		
1	1	0							1	
1	1	1								1

E	A ₀	A ₁	A ₂	D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇
0	x	x	x	0	0	0	0	0	0	0	0
1	0	0	0	1							
1	0	0	1		1						
1	0	1	0			1					
1	0	1	1				1				
1	1	0	0					1			
1	1	0	1						1		
1	1	1	0							1	
1	1	1	1								1

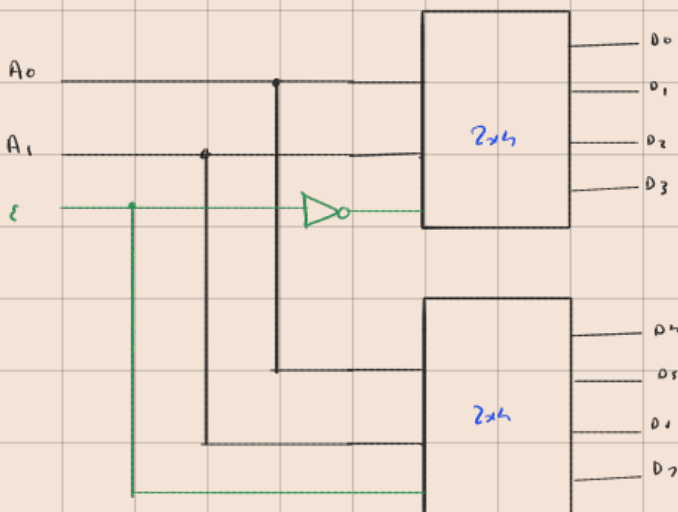
Active Low

E	A ₀	A ₁	A ₂	D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇
0	x	x	x	0	0	0	0	0	0	0	0
1	0	0	0	1							
1	0	0	1		1						
1	0	1	0			1					
1	0	1	1				1				
1	1	0	0					1			
1	1	0	1						1		
1	1	1	0							1	
1	1	1	1								1

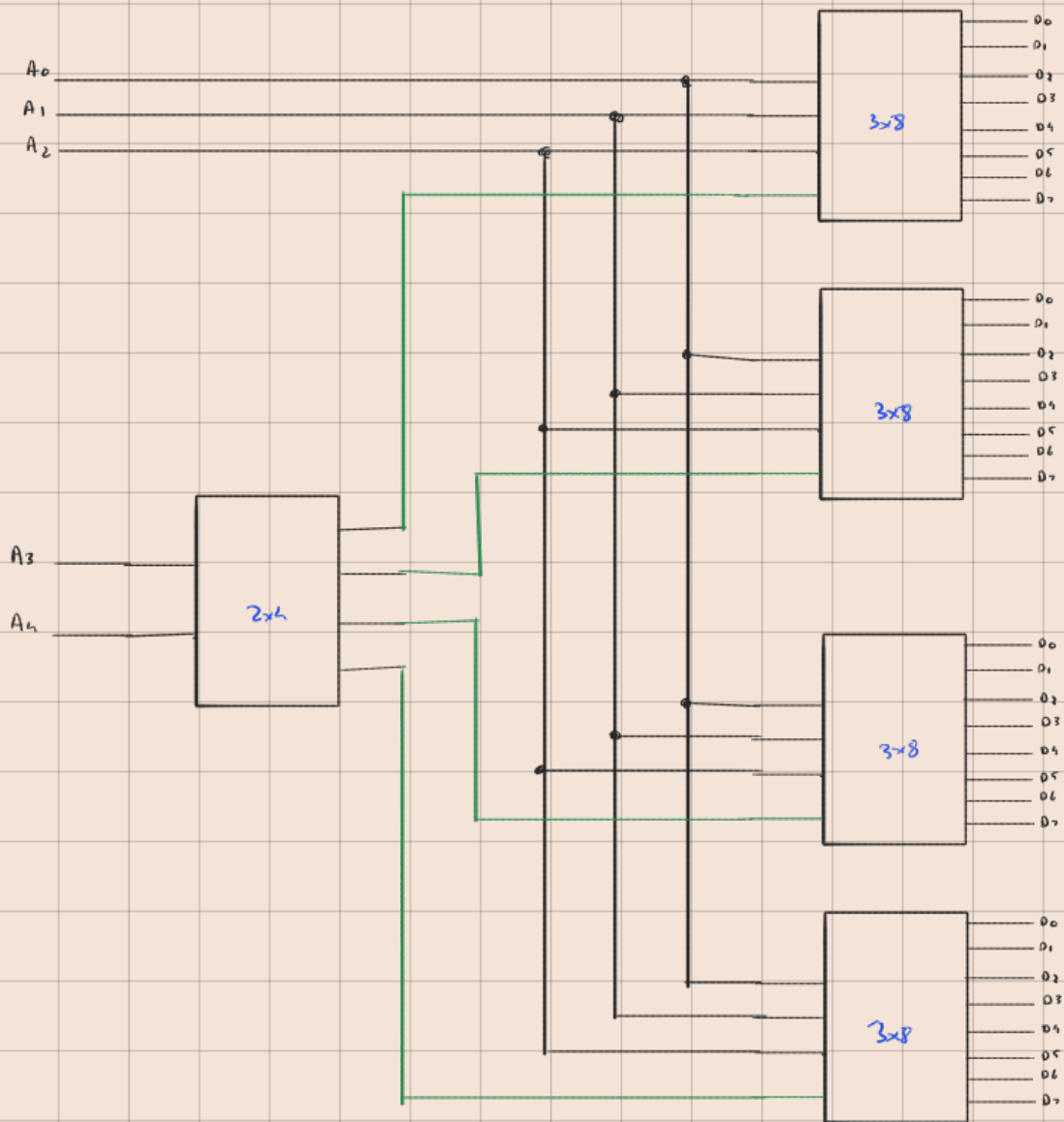
4x16 Decoder Using 3x8 Decoder



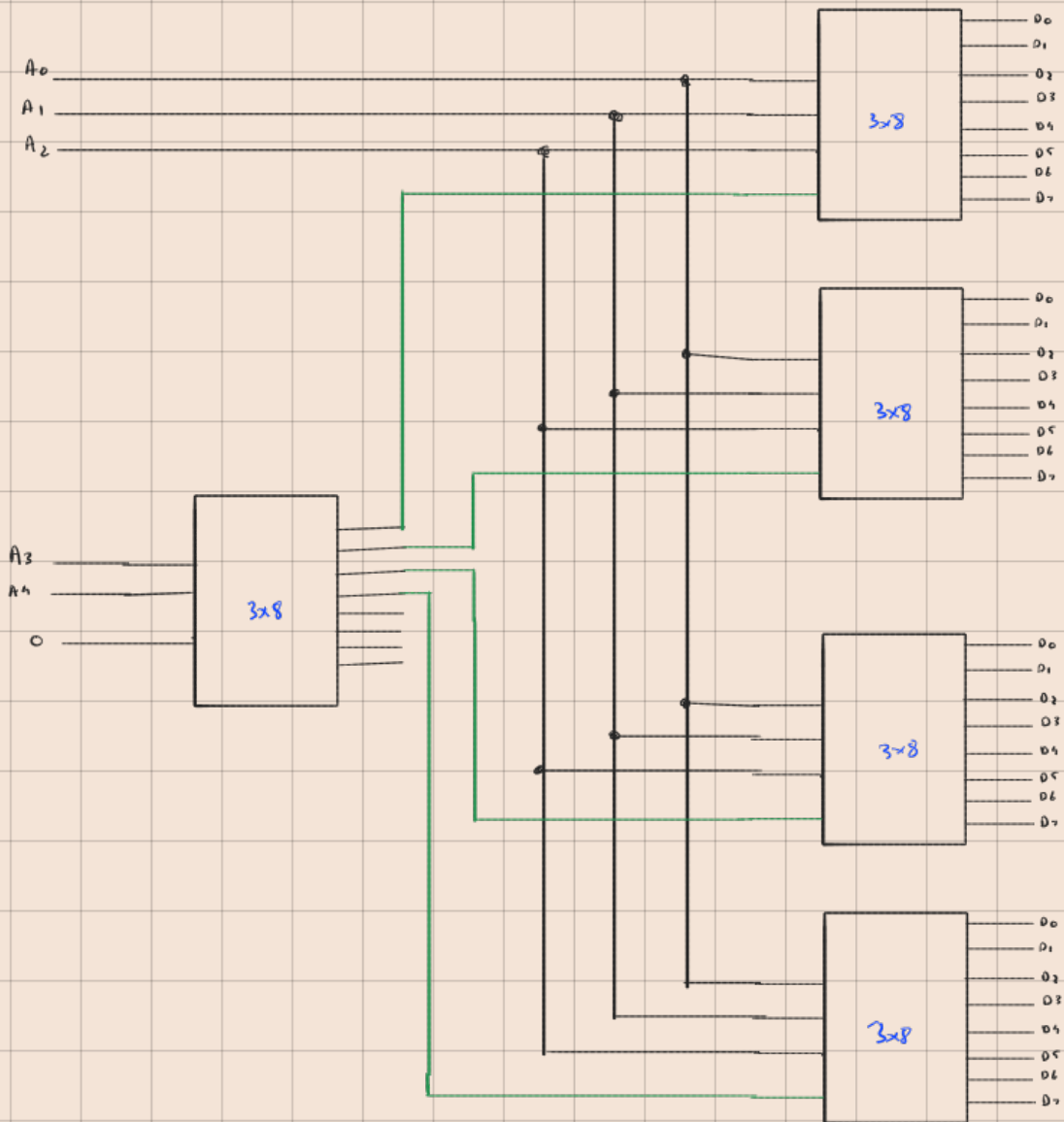
3x8 Decoder Using 2x4 Decoders



S = 12 Decoder Using 3x8



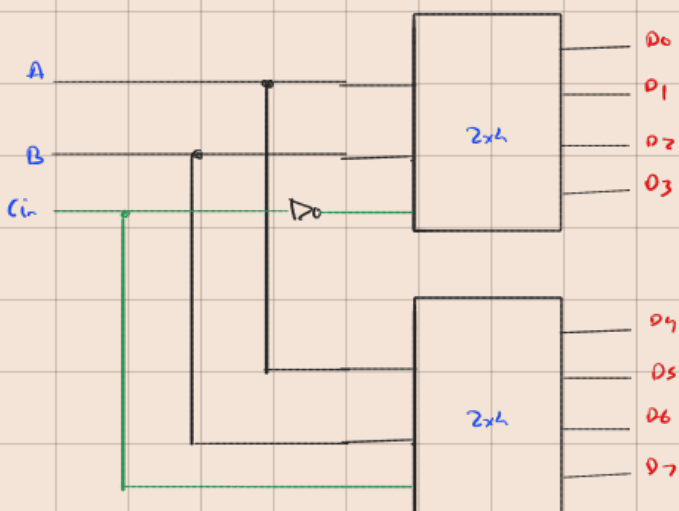
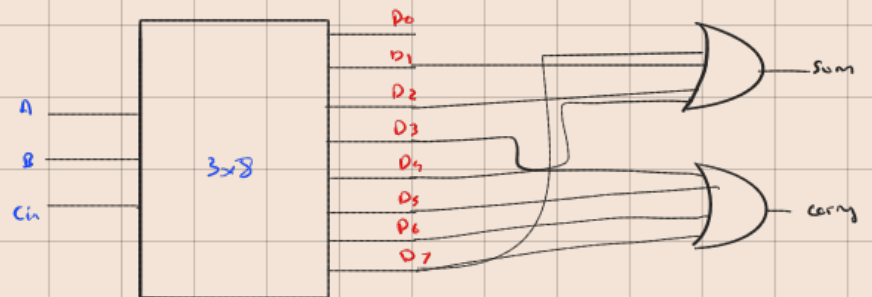
5x32 Decoder Using 8x8 (Only 3x8)



Logic Circuit Implementation

Full Adder Using 2x4

A	B	Cin	Sum	Cout	
0	0	0	0	0	D ₀
0	0	1	1	0	D ₁
0	1	0	1	0	D ₂
0	1	1	0	1	D ₃
1	0	0	1	0	D ₄
1	0	1	0	1	D ₅
1	1	0	0	1	D ₆
1	1	1	1	1	D ₇



Decoder with Active Low output

Implementing Logic Expression Using Decoder

1. SOP (Sum of Products) Implementation

Given:

$$F = \Sigma m(1, 3, 5, 7) \quad F = \Pi M(0, 2, 4, 6)$$

Expanded Form:

$$F = \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}C + ABC$$

Using 3×8 Decoder:

- Inputs: A, B, C
- Outputs: D_0 to D_7
- Select outputs: D_1, D_3, D_5, D_7

Final Implementation:

$$F = D_1 + D_3 + D_5 + D_7$$

	A	B	C	D_0	$\bar{D}_1$	D_2	$\bar{D}_3$	D_4	$\bar{D}_5$	D_6	$\bar{D}_7$
0	0	0	0	0	1	0	0	0	0	0	1
1	0	0	1	0	0	1	0	0	0	0	0
2	0	1	0	0	1	0	1	0	0	0	0
3	0	1	1	0	1	1	0	0	0	0	0
4	1	0	0	1	0	0	1	0	0	0	0
5	1	0	1	1	0	1	0	1	0	0	0
6	1	1	0	1	0	1	1	0	0	1	0
7	1	1	1	1	0	1	1	0	0	1	0

